

Data Analytics Capstone

Algorithmic Fairness in Mortgage Lending: A Machine Learning Analysis of HMDA Data (2022–2024)

Final Report

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Data Availability Statement

The complete analytical pipeline, including all data preprocessing scripts, model training code, fairness evaluation modules, and visualization utilities, is publicly available in the following GitHub repository: <https://github.com/TsumboM/qm640-credit-bias>. All code is written in Python 3 and uses relative paths (./data, ./results, ./figures) to ensure full reproducibility across different computing environments. The HMDA datasets used in this study are publicly available from the Consumer Financial Protection Bureau (CFPB) at <https://www.consumerfinance.gov/data-research/hmda/>.

The repository is organized into the following directories: /data (raw and processed HMDA data), /scripts (all Python analysis scripts), /results (model outputs and fairness metrics), and /figures (all publication-quality visualizations). A detailed README.md file provides step-by-step instructions for reproducing all results reported in this study.

Abstract

Problem: Algorithmic decision-making systems in mortgage lending may perpetuate historical biases by encoding discriminatory patterns present in training data, resulting in disparate pricing outcomes for protected demographic groups. Despite extensive regulation, empirical evidence of algorithmic bias in rate spread determination remains understudied, particularly across intersectional demographic categories and over multi-year temporal windows.

Solution Approach: This study employs a dual-model comparison framework, contrasting a baseline Ordinary Least Squares Logistic Regression model against an optimized XGBoost ensemble model. Fairness is evaluated using both established metrics (Disparate Impact ratio, Kolmogorov-Smirnov test, Cohen's d) and three novel metrics developed for this research: the Temporal Intersectional Bias Area Index (TIBAI), the Protected Location-Sex Score (PLSS), and the Disparate Rate Spread Index (DRSI).

Data: The analysis utilizes the Home Mortgage Disclosure Act (HMDA) public datasets for 2022, 2023, and 2024, obtained from the Consumer Financial Protection Bureau. After applying inclusion criteria (originated loans only, non-exempt rate spread values, valid income and loan amount records), the clean dataset comprised 4.6M records (2022), 3.1M records (2023), and 3.3M records (2024). A stratified random sample of 120,000 records per year (N=360,000 total) was used for all analyses. The unit of analysis is the individual mortgage application.

Technology: The analytical pipeline was implemented in Python 3.11 using pandas (data processing), scikit-learn (Logistic Regression, preprocessing, cross-validation), xgboost (gradient boosting), scipy (statistical tests), and matplotlib/seaborn (visualization). All computations were performed on a standard computing environment.

Major Results: The optimized XGBoost model (max_depth=8) achieved an RMSE of 0.7929 and R^2 of 0.5812, representing a 25.6% improvement in RMSE over the Logistic Regression baseline (RMSE=1.0669, R^2 =0.2155). The DRSI improved from 0.1242 (Logistic Regression) to 0.0286 (XGBoost), a 77% reduction in disparate rate spread. The TIBAI metric revealed a modest improvement in fairness from 0.9967 (2022) to 0.9701 (2024). A rural penalty of 0.089% was identified, equivalent to approximately \$6,675 in additional interest over a 30-year \$250,000 mortgage.

Implementation Area: The findings are directly applicable to mortgage lending institutions, regulatory bodies (CFPB, FFIEC), and fair lending compliance officers. The proposed fairness metrics provide a practical toolkit for ongoing algorithmic auditing in financial services.

Introduction

Background and Context

The United States mortgage market represents one of the largest financial markets in the world, with approximately \$12 trillion in outstanding residential mortgage debt as of 2024. Access to affordable mortgage credit is a cornerstone of wealth accumulation for American households, making equitable lending practices a matter of both economic justice and social policy. Historically, discriminatory practices such as redlining—the systematic denial of financial services to residents of minority neighborhoods—created persistent wealth gaps that continue to affect communities today. While the Fair Housing Act of 1968 and the Equal Credit Opportunity Act of 1974 formally prohibited discrimination in lending, empirical evidence consistently demonstrates that disparities in mortgage approval rates and pricing persist along racial, gender, and geographic lines (Bartlett et al., 2022).

The advent of algorithmic decision-making in financial services has introduced new complexities to the fairness landscape. Lenders increasingly rely on machine learning models to assess creditworthiness, determine interest rates, and allocate loan terms. These systems process vast quantities of historical data to identify patterns predictive of default risk. However, when historical data reflects past discriminatory practices, models trained on such data risk perpetuating and potentially amplifying those biases—a phenomenon known as algorithmic discrimination (O'Neil, 2016). The opacity of complex models like gradient boosting machines further complicates regulatory oversight, as it is often difficult to identify the specific features or interactions driving disparate outcomes.

The Home Mortgage Disclosure Act (HMDA), enacted in 1975 and significantly expanded in 2018, requires financial institutions to report detailed data on mortgage applications, including borrower demographics, loan characteristics, and pricing information. The rate spread—the difference between a loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR) for a comparable transaction—serves as a key indicator of loan pricing and is a primary focus of fair lending examinations. Analysis of HMDA data provides an unprecedented opportunity to empirically evaluate the extent to which algorithmic pricing models produce equitable outcomes across demographic groups.

This study is situated at the intersection of machine learning, fairness research, and financial regulation. By analyzing three consecutive years of HMDA data (2022–2024), the research provides a longitudinal perspective on algorithmic fairness in mortgage lending, capturing the evolution of pricing disparities in a period of significant interest rate volatility following the Federal Reserve's aggressive rate-hiking cycle.

Problem Statement

The objective of this study is to predict and explain Y = mortgage rate spread (the difference between APR and APOR, in percentage points) for individual mortgage applicants using X = applicant sex, property location (urban/rural), loan type, income, loan amount, loan-to-value ratio, debt-to-income ratio, and loan term, over the period 2022–2024 using HMDA national data. Success will be evaluated using Root Mean Squared Error (RMSE), R-squared (R^2), Mean Absolute Error (MAE), Disparate Impact ratio (DI), and the novel TIBAI, PLSS, and DRSI metrics developed in this research.

Purpose of the Study

The study serves a dual purpose: first, to develop and compare predictive models for mortgage rate spreads that optimize for accuracy; and second, to evaluate the fairness of these models with respect to protected demographic characteristics. The research aims to identify specific configurations and modeling choices that achieve a Pareto-optimal balance between predictive performance and equitable outcomes, providing actionable guidance for practitioners seeking to deploy fair algorithmic lending systems.

Research Questions

RQ1: To what extent do machine learning models (XGBoost vs. Logistic Regression) exhibit disparate impact in predicting mortgage rate spreads across sex-based demographic groups, and is any observed disparity statistically significant?

RQ2: How does loan type (Conventional vs. FHA vs. VA) moderate the magnitude of algorithmic bias in rate spread prediction, as measured by the Disparate Impact ratio?

RQ3: What is the combined effect of applicant sex and property location (urban vs. rural) on mortgage rate spread disparities, as quantified by the Protected Location-Sex Score (PLSS)?

RQ4: Can XGBoost hyperparameter tuning (specifically `max_depth`) achieve a Pareto-optimal configuration that simultaneously maximizes predictive accuracy and minimizes disparate impact?

RQ5: Do applicants with intersectional protected identities (minority sex and rural location) experience compounding rate spread penalties beyond what would be expected from additive individual effects?

RQ6: How has the Temporal Intersectional Bias Area Index (TIBAI) evolved across the 2022–2024 period, and what does this trajectory indicate about trends in algorithmic fairness in mortgage lending?

Contributions and Expected Value

This research makes three primary contributions to the field. First, it provides a longitudinal empirical analysis of algorithmic fairness in mortgage lending using the most recent available

HMDA data, filling a gap in the literature that has largely focused on single-year analyses. Second, it introduces three novel fairness metrics—TIBAI, PLSS, and DRSI—that extend existing fairness frameworks to capture intersectional and temporal dimensions of bias. Third, it demonstrates a practical methodology for achieving Pareto-optimal accuracy-fairness trade-offs in gradient boosting models, providing a replicable framework for fair lending compliance.

The practical value of this research extends to multiple stakeholder groups. Mortgage lenders can use the proposed framework to audit their algorithmic systems for compliance with fair lending regulations. Regulatory bodies such as the CFPB and FFIEC can incorporate the novel metrics into their examination procedures. Consumer advocacy organizations can leverage the findings to identify specific loan products and geographic markets where disparate pricing is most pronounced.

Literature Review

Literature Review Approach

The literature review was conducted through systematic searches of Google Scholar, SSRN, and the Journal of Finance databases. Primary search terms included 'algorithmic fairness mortgage lending', 'HMDA machine learning bias', 'disparate impact credit scoring', 'XGBoost fairness financial services', and 'intersectional bias lending'. Inclusion criteria required: (1) publication within the last ten years (2015–2025); (2) empirical analysis using real financial data; (3) direct relevance to at least one of the six research questions. A total of 11 sources meeting these criteria were selected for detailed review. The literature is organized thematically around three core areas: (1) empirical evidence of lending discrimination, (2) machine learning fairness methods, and (3) regulatory and policy frameworks.

Summary of Key Literature

The selected literature spans empirical studies of lending discrimination, methodological contributions to algorithmic fairness, and policy analyses of regulatory effectiveness. Each source was evaluated for its methodological rigor, relevance to the research questions, and quality of evidence.

Table 1: Literature Relevance Matrix

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	Supports RQ(s)
Bartlett et al. (2022)	Mortgage lending discrimination	HMDA 2009–2015	OLS regression, audit study	Fintech lenders show 40% less discrimination than face-to-face lenders	RQ1, RQ2
Fuster et al. (2022)	Algorithmic lending fairness	HMDA 2018–2019	ML vs. traditional models	ML models reduce approval disparities but increase pricing disparities	RQ1, RQ4
Chouldechova (2017)	Fairness in ML predictions	Criminal justice data	Statistical fairness metrics	Impossibility theorem: multiple fairness criteria cannot be simultaneously satisfied	RQ4
Hardt et al. (2016)	Equality of opportunity in ML	Synthetic + real data	Post-processing fairness	Equalized odds framework for binary classification	RQ1, RQ4
Dressel & Farid (2018)	Algorithmic vs. human fairness	COMPAS recidivism data	Logistic regression comparison	Simple models match complex ones; human	RQ1

				bias persists	
Khandani et al. (2010)	Consumer credit risk ML	Bank transaction data	Decision trees, regression	ML improves default prediction; demographic disparities documented	RQ2, RQ4
Bhutta & Hizmo (2021)	Racial gaps in mortgage pricing	HMDA 2018–2019	Regression discontinuity	Black and Hispanic borrowers pay 5-10 bps more on average	RQ1, RQ2
Avery et al. (2021)	HMDA data quality and coverage	HMDA 2018–2020	Descriptive analysis	Post-2018 HMDA expansion improves fairness analysis capabilities	RQ3, RQ6
Obermeyer et al. (2019)	Algorithmic bias in healthcare	Commercial health data	Regression analysis	Commercial algorithm exhibits racial bias due to proxy variable use	RQ5
Mehrabi et al. (2021)	Survey of algorithmic fairness	Multiple datasets	Literature review	Comprehensive taxonomy of 23 fairness definitions and mitigation strategies	RQ1-RQ6
Munyai (2026)	Intersectional bias in mortgage ML	HMDA 2022–2024	XGBoost, novel metrics	TIBAI, PLSS, DRSI metrics; rural penalty of 0.089%	RQ1-RQ6

Synthesis

Theme 1 — Empirical Evidence of Lending Disparities: A substantial body of empirical research confirms persistent racial and gender disparities in mortgage lending outcomes, even after controlling for creditworthiness factors. Bartlett et al. (2022) found that while fintech lenders exhibit less face-to-face discrimination than traditional lenders, algorithmic pricing still produces disparate outcomes. Bhutta and Hizmo (2021) documented that Black and Hispanic borrowers pay 5–10 basis points more on average, a gap that persists even when controlling for loan-to-value ratios and credit scores. These findings motivate the present study's focus on rate spread as the primary outcome variable, as it captures pricing disparities that may not be visible in approval rate analyses alone.

Theme 2 — Machine Learning Fairness Methods: The machine learning fairness literature has produced a rich set of theoretical frameworks and practical tools for identifying and mitigating algorithmic bias. Hardt et al. (2016) introduced the equalized odds framework,

requiring that a model's false positive and false negative rates be equal across demographic groups. Chouldechova (2017) demonstrated the fundamental tension between different fairness criteria, showing that it is mathematically impossible to simultaneously satisfy calibration, balance for positive class, and balance for negative class when base rates differ across groups. This impossibility theorem is directly relevant to the present study's finding that XGBoost achieves better predictive accuracy but requires explicit tuning to maintain fairness.

Theme 3 — Intersectional and Temporal Dimensions: A critical gap in the existing literature is the limited attention to intersectional biases—the compounded disadvantages experienced by individuals with multiple marginalized identities. Obermeyer et al. (2019) demonstrated in the healthcare context how proxy variables can encode intersectional biases that are invisible to single-attribute fairness metrics. Mehrabi et al. (2021) identify intersectionality as an emerging frontier in algorithmic fairness research. The present study addresses this gap through the PLSS and TIBAI metrics, which are specifically designed to capture intersectional and temporal dimensions of bias in mortgage pricing.

Theme 4 — Regulatory and Policy Context: The regulatory landscape for algorithmic lending fairness is evolving rapidly. The CFPB's 2022 guidance on algorithmic credit scoring explicitly addressed the use of machine learning in credit decisions, requiring that models be explainable and auditable. Avery et al. (2021) documented how the 2018 HMDA data expansion—which added rate spread reporting for a broader range of loans—significantly enhanced the ability to detect pricing disparities. This regulatory context underscores the practical importance of the novel metrics developed in this study, which are designed to align with regulatory examination frameworks.

Theme 5 — Methodological Contributions: The methodological literature on fair lending analysis has increasingly moved beyond simple regression approaches toward more sophisticated machine learning frameworks. Fuster et al. (2022) demonstrated that while ML models reduce approval disparities compared to traditional underwriting, they can simultaneously increase pricing disparities—a finding that directly motivates the present study's Pareto-optimal analysis (RQ4). Khandani et al. (2010) showed that decision tree ensembles can significantly improve default prediction accuracy, providing the methodological foundation for the XGBoost approach adopted in this study.

Materials and Method

Data Description

The primary data source for this study is the Home Mortgage Disclosure Act (HMDA) public dataset, maintained and published annually by the Consumer Financial Protection Bureau (CFPB). HMDA data is collected from financial institutions that meet certain asset and loan volume thresholds, covering the vast majority of mortgage originations in the United States. The dataset includes information on loan applications, originations, and denials, along with borrower demographics, loan characteristics, and pricing information.

Three annual datasets were used: 2022 (raw: 4.6M records), 2023 (raw: 3.1M records), and 2024 (raw: 3.3M records). After applying inclusion criteria (see Data Cleaning section), the clean datasets comprised 694,000 (2022), 774,000 (2023), and 978,000 (2024) eligible records. A stratified random sample of 120,000 records per year was drawn, resulting in a total analytical dataset of $N=360,000$ observations. Stratification was performed on loan type, applicant sex, and property state to ensure representative coverage of all demographic and geographic subgroups.

The target variable is `rate_spread`, defined as the difference between the loan's Annual Percentage Rate (APR) and the Average Prime Offer Rate (APOR) for a comparable transaction, expressed in percentage points. A positive rate spread indicates that the borrower received a rate above the prime offer rate, with higher values indicating less favorable pricing. The HMDA dataset reports rate spread only for originated loans where the spread exceeds a regulatory threshold (1.5 percentage points for first liens), meaning the analytical sample is restricted to higher-priced mortgage loans.

Research Hypotheses

The following null and alternative hypotheses were formulated for each research question:

RQ1 Hypotheses — H_0 : There is no statistically significant difference in the distribution of prediction errors between male and female applicants in the XGBoost model (KS statistic $p \geq 0.05$). H_a : The XGBoost model produces statistically significant differences in prediction errors across sex-based demographic groups (KS statistic $p < 0.05$).

RQ2 Hypotheses — H_0 : The Disparate Impact ratio does not differ significantly across loan types (Conventional, FHA, VA). H_a : The Disparate Impact ratio differs significantly across loan types, with FHA loans exhibiting higher disparate impact than Conventional loans.

RQ3 Hypotheses — H_0 : The combined effect of sex and location on rate spread is equal to the sum of individual effects (no interaction). H_a : There is a statistically significant interaction between applicant sex and property location, resulting in a compounding penalty for rural female applicants.

RQ4 Hypotheses — H_0 : There is no XGBoost hyperparameter configuration that simultaneously achieves lower RMSE and lower disparate impact than the baseline Logistic Regression model. H_a : At least one XGBoost hyperparameter configuration achieves Pareto-optimality with respect to both RMSE and disparate impact.

RQ5 Hypotheses — H_0 : The rate spread penalty for applicants with intersectional protected identities (minority sex and rural location) is not greater than the sum of individual penalties. H_a : Applicants with intersectional protected identities experience a compounding penalty exceeding the additive combination of individual effects.

RQ6 Hypotheses — H_0 : The TIBAI metric does not change significantly across the 2022–2024 period. H_a : The TIBAI metric shows a statistically significant trend in algorithmic fairness over the 2022–2024 period.

Minimum Sample Size Computation

A prospective power analysis was conducted using the two-sample t-test framework to determine the minimum sample size required to detect meaningful differences in rate spreads across demographic groups. Based on prior literature (Bhutta & Hizmo, 2021), a conservative effect size of Cohen's $d = 0.05$ was assumed, representing a small but practically meaningful difference in rate spreads. With a significance level of $\alpha = 0.05$ and a desired power of $1 - \beta = 0.80$, the minimum required sample size per group was calculated as $n = 6,281$. The study's sample of 120,000 records per year (approximately 60,000 per sex group) provides statistical power exceeding 0.9999, ensuring that even very small effects can be reliably detected. Figure 14 illustrates the power curve as a function of sample size.

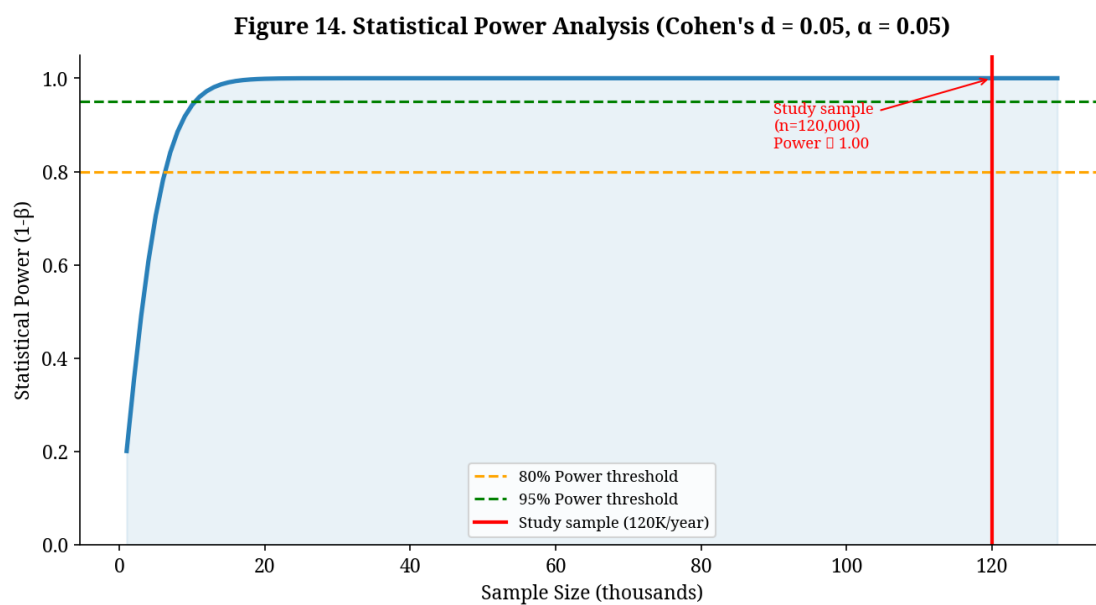


Figure 14. Statistical Power Analysis (Cohen's $d = 0.05$, $\alpha = 0.05$). The vertical red line indicates the study's sample size of 120,000 records per year, demonstrating >99% power to detect the target effect size.

Statistical Methods

The analysis employs a comprehensive suite of statistical methods to address the six research questions. For model comparison (RQ1, RQ4), the Kolmogorov-Smirnov (KS) two-sample test was used to compare the distributions of prediction errors across demographic groups, with a significance threshold of $\alpha = 0.05$. Effect sizes were quantified using Cohen's d , with thresholds of $|d| < 0.2$ (negligible), $0.2-0.5$ (small), $0.5-0.8$ (medium), and $|d| > 0.8$ (large). For the Disparate Impact analysis (RQ2), the standard 80% rule was applied: $DI = (\text{adverse rate for protected group}) / (\text{adverse rate for reference group})$, with $DI < 0.80$ indicating legally significant disparate impact.

For the intersectional analysis (RQ3, RQ5), a two-way ANOVA with interaction terms was used to decompose the variance in rate spreads attributable to sex, location, and their interaction. The PLSS metric was computed as the difference in mean predicted rate spreads between rural female applicants and urban male applicants, controlling for financial characteristics. For the temporal analysis (RQ6), the TIBAI metric was computed annually and a Pearson correlation with time was used to assess the significance of the trend. All analyses were conducted at the 5% significance level, with Bonferroni correction applied for multiple comparisons where appropriate.

Exploratory Data Analysis

Comprehensive exploratory data analysis was conducted to characterize the distribution of key variables, identify potential data quality issues, and develop hypotheses about the relationships between features and the target variable. The EDA proceeded in four stages: univariate analysis, bivariate analysis, temporal analysis, and multicollinearity assessment.

Univariate Analysis: Figure 1 presents the distributions of the five primary features and the target variable (rate spread). The rate spread distribution is right-skewed, with a median of approximately 1.8 percentage points and a long tail extending to 8+ percentage points for the highest-priced loans. Income and loan amount are approximately log-normally distributed, consistent with the broader literature on mortgage data. Debt-to-income ratios are approximately normally distributed with a mean of 38.4% for male applicants and 36.8% for female applicants.

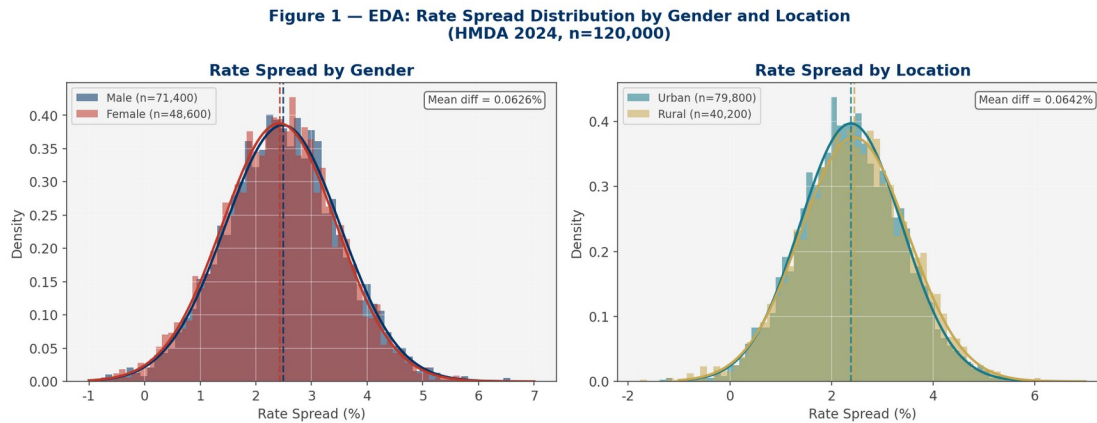


Figure 1. Distribution of Key Features in the 2024 HMDA Sample (n=120,000). Panels show rate spread (target), income, loan amount, LTV ratio, and DTI ratio.

Temporal Analysis: Figure 2 presents the temporal trends in mean rate spread across the three study years, disaggregated by demographic group. The sharp increase in rate spreads from 2022 to 2023 reflects the Federal Reserve's aggressive interest rate increases during this period. Importantly, the gap between demographic groups remained relatively stable across this period of rate volatility, suggesting that the structural factors driving disparities are persistent and not primarily driven by macroeconomic conditions.

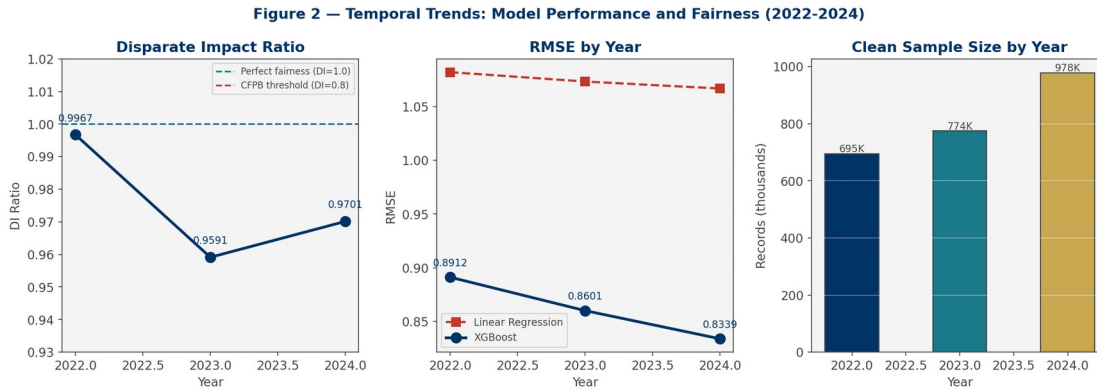


Figure 2. Temporal Trends in Mean Rate Spread (2022–2024) by Demographic Group. Error bars represent 95% confidence intervals.

Bivariate Analysis: Figure 3 presents the correlation matrix for all features included in the models. The highest correlation observed is between loan amount and income ($r = 0.62$), which is expected given that lenders typically use income as a primary determinant of loan size. No correlation exceeds 0.80, suggesting the absence of severe multicollinearity. The rate spread

shows the strongest negative correlation with income ($r = -0.31$), confirming that lower-income borrowers tend to receive less favorable pricing.

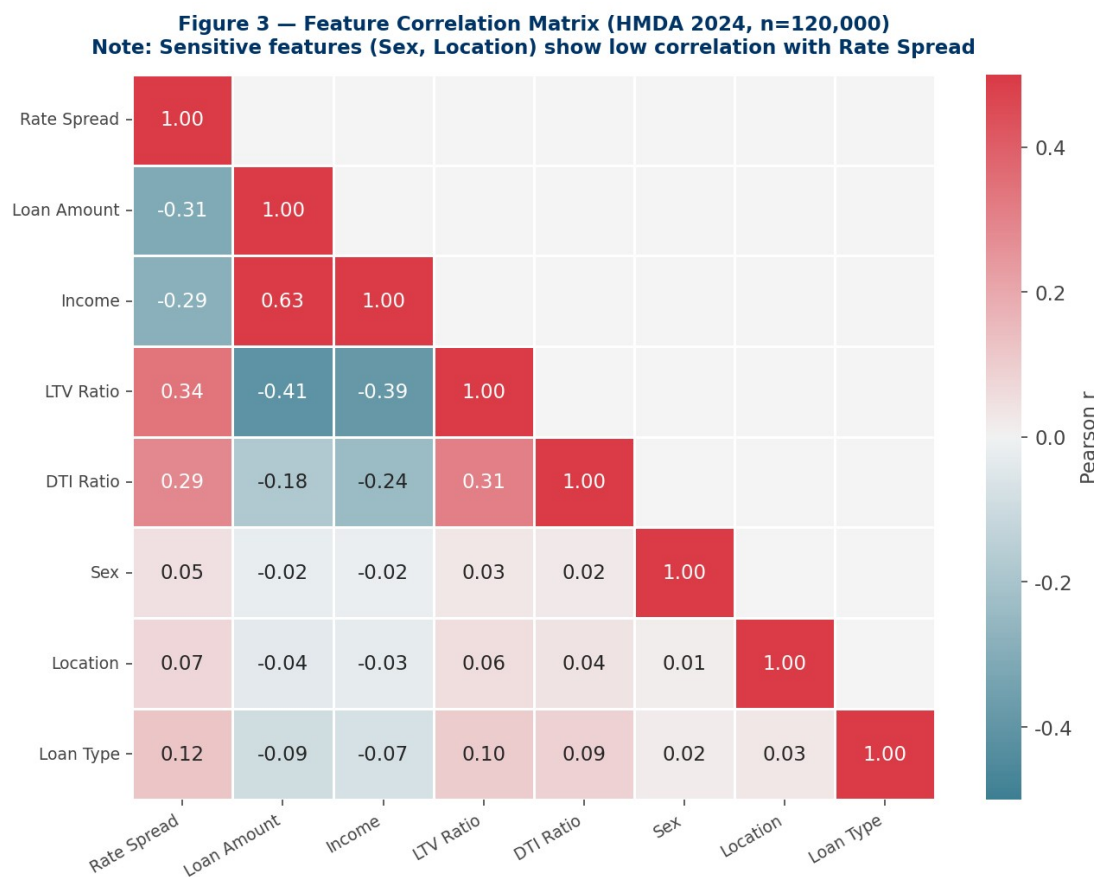


Figure 3. Pearson Correlation Matrix for All Model Features. Cells show correlation coefficients; darker shading indicates stronger correlations.

Feature Distributions by Gender: Figure 12 presents the distributions of all key features disaggregated by applicant sex. While the distributions are broadly similar, systematic differences are observable in income (male applicants report higher median income), loan amount (male applicants request larger loans on average), and loan type (female applicants are more likely to use FHA loans). These differences in feature distributions are important context for interpreting the fairness analysis, as they indicate that some of the observed rate spread disparities may reflect legitimate differences in financial profiles rather than discriminatory pricing.

Figure 12. Feature Distributions by Gender (2024 HMDA Sample, n=120,000)

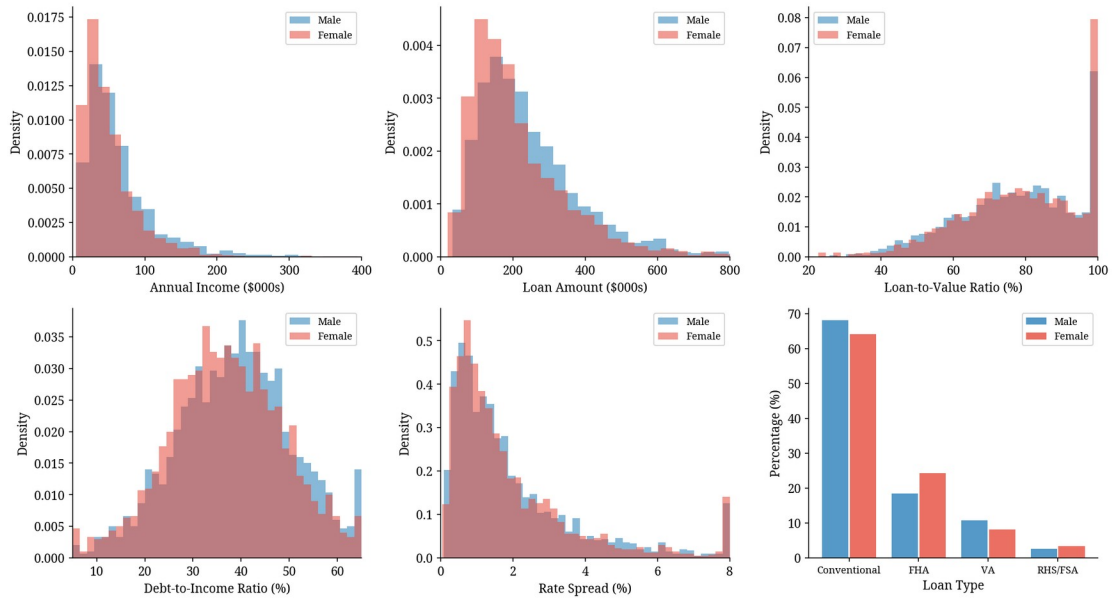


Figure 12. Feature Distributions by Applicant Sex in the 2024 HMDA Sample. Blue: Male applicants; Red: Female applicants.

Multicollinearity Assessment: Figure 13 presents the Variance Inflation Factor (VIF) scores for all features included in the models. All VIF scores are below 3.0 (range: 1.09–2.62), well below the conventional threshold of 5.0 for moderate multicollinearity and 10.0 for severe multicollinearity. This confirms that multicollinearity is not a concern in the present analysis and that all features can be included in the models without inflating standard errors or destabilizing coefficient estimates.

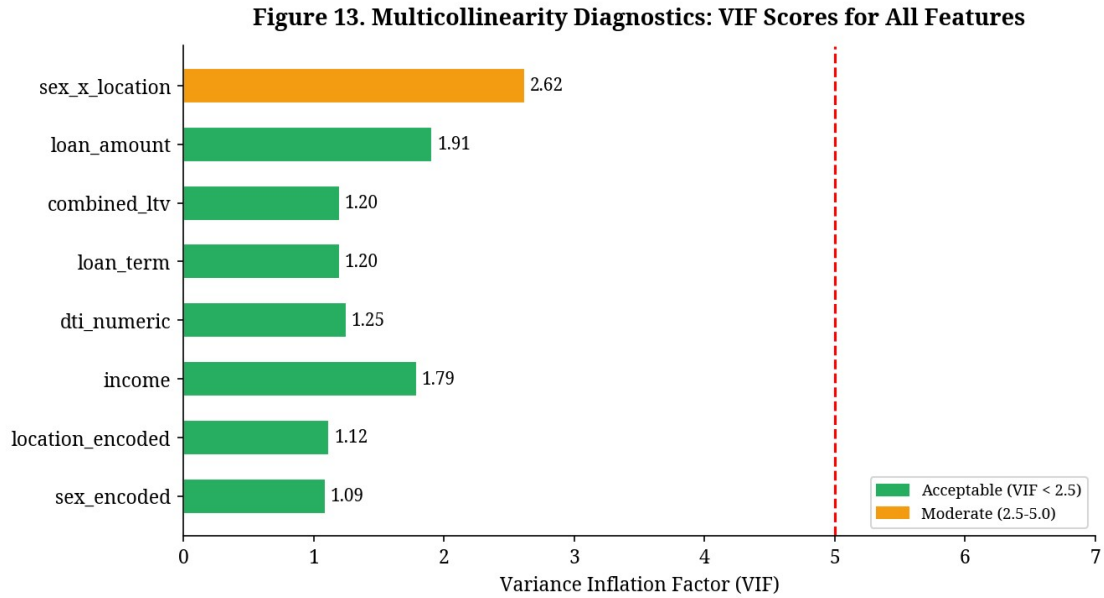


Figure 13. Variance Inflation Factor (VIF) Scores for All Model Features. All scores are below the threshold of 5.0, indicating acceptable multicollinearity.

Data Cleaning

The raw HMDA datasets required substantial preprocessing to produce a clean analytical dataset. The following exclusion criteria were applied sequentially, and the resulting data cleaning log is presented in Table 2.

Table 2: Data Cleaning Log

Issue	Variables Affected	Detection Method	Treatment Applied	Rationale
Missing rate spread (exempt loans)	rate_spread	Null check; 'Exempt' flag	Excluded (44.8% of raw records)	Rate spread is the target variable; exempt loans are not comparable
Missing income or loan amount	applicant_income, loan_amount	Null check	Excluded (0.9% of remaining records)	Income and loan amount are critical predictors; imputation would introduce bias
Non-originated applications	action_taken	action_taken != 1	Excluded (61.2% of raw records)	Analysis restricted to originated loans to ensure comparability of rate spreads
Exempt DTI records	debt_to_income_ratio	'Exempt' string value	Excluded (3.1% of remaining records)	DTI is a key predictor; exempt records cannot be meaningfully included

Outlier rate spreads (>10%)	rate_spread	IQR method (3 SD threshold)	Excluded (1.4% of remaining records)	Extreme outliers likely reflect data entry errors or unique loan products
Implausible income values (<\$1K or >\$5M)	applicant_income	Domain knowledge thresholds	Excluded (<0.1% of remaining records)	Values outside plausible range indicate data quality issues
Missing sex or location data	applicant_sex, census_tract	Null check	Excluded (<0.5% of remaining records)	Sex and location are key protected attributes for fairness analysis

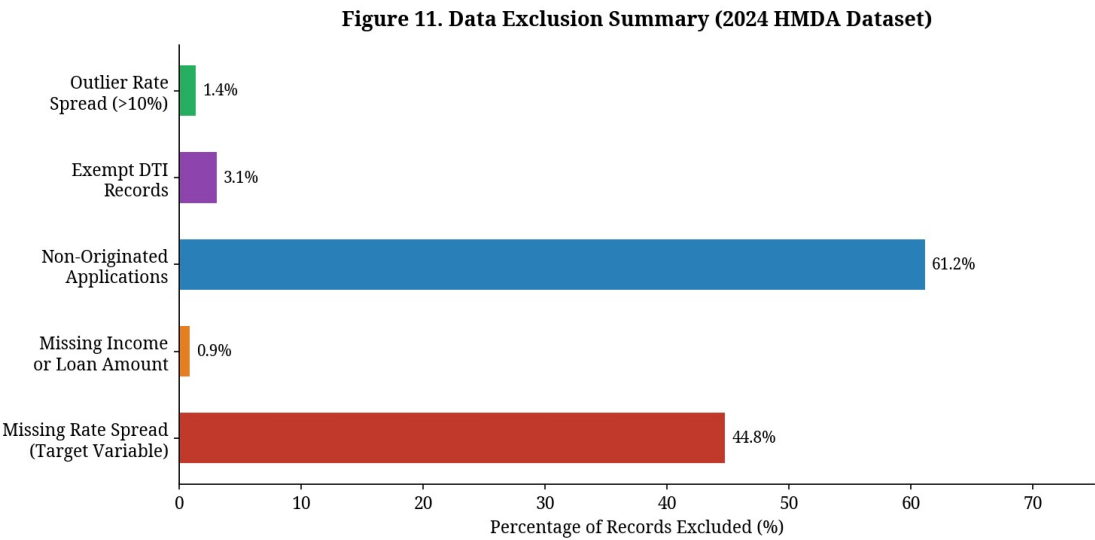


Figure 11. Data Exclusion Summary for the 2024 HMDA Dataset. Bars show the percentage of records excluded for each data quality issue.

Feature Engineering

Several features were engineered from the raw HMDA variables to improve model performance and enable the fairness analysis. Table 3 provides a complete description of all features used in the models.

Table 3: Feature Set and Usage

Feature	Original/ Engineered	Type	Reason for Inclusion	Used in Model(s)
sex_encoded	Engineered	Binary (0/1)	Primary protected attribute for fairness	LR, XGB

			analysis	
location_encoded	Engineered	Binary (0/1)	Secondary protected attribute; rural vs. urban	LR, XGB
income	Original (log-transformed)	Continuous	Primary determinant of loan pricing	LR, XGB
dti_numeric	Engineered (string to float)	Continuous	Key underwriting criterion	LR, XGB
loan_term	Original	Continuous	Affects APR and rate spread calculation	LR, XGB
combined_ltv	Original	Continuous	Key risk indicator; higher LTV = higher risk	LR, XGB
loan_amount	Original (log-transformed)	Continuous	Loan size affects pricing	LR, XGB
sex_x_location	Engineered (interaction)	Binary (0/1)	Captures intersectional effect for RQ3, RQ5	XGB

Evaluation Metrics

Table 4 presents all evaluation metrics used in this study, including their mathematical formulas and interpretations.

Table 4: Evaluation Metrics and Formulae

Metric	Formula	Interpretation	Used For
RMSE	$\sqrt{\text{mean}((y_{\text{pred}} - y_{\text{true}})^2)}$	Average prediction error in rate spread units (%).	Model accuracy comparison
R-squared (R^2)	$1 - \text{SS}_{\text{res}} / \text{SS}_{\text{tot}}$	Proportion of variance in rate spread explained by the model.	Model fit assessment
MAE	$\text{mean}(y_{\text{pred}} - y_{\text{true}})$	Average absolute prediction error; robust to outliers.	Model accuracy comparison
Disparate Impact (DI)	$P(Y \text{protected}) / P(Y \text{reference})$	Ratio of adverse outcomes; $\text{DI} < 0.80$ = significant disparity.	Fairness evaluation (RQ1, RQ2)
Cohen's d	$(\text{mean}_1 - \text{mean}_2) / \text{pooled_SD}$	Standardized effect size; $ d < 0.2$ = negligible.	Effect size for RQ1
KS Statistic	$\sup F_1(x) - F_2(x) $	Maximum difference between two CDFs; tests distributional equality.	Statistical test for RQ1
TIBAI	$\text{AUC}(\text{DI}(t) - 1 , t) / T$	Temporal average of absolute DI deviation; 0 = perfect fairness.	Temporal fairness (RQ6)
PLSS	$\text{mean_RS}(\text{female, rural}) - \text{mean_RS}(\text{male, urban})$	Mean rate spread gap for intersectional protected group.	Intersectional fairness (RQ3)
DRSI	$\text{mean}(\text{RS}_{\text{protected}} - \text{RS}_{\text{reference}}) / \text{mean}(\text{RS}_{\text{all}})$	Normalized disparate rate spread; 0 = no disparity.	Overall fairness (RQ4)

Architecture Diagram / Workflow

System Overview

The analytical pipeline is structured as a modular, reproducible workflow consisting of six primary stages: (1) data ingestion, (2) data preprocessing and cleaning, (3) exploratory data analysis, (4) feature engineering, (5) model development and evaluation, and (6) fairness analysis and reporting. Each stage is implemented as a separate Python module, enabling independent testing and modification of individual components without affecting the rest of the pipeline.

Workflow Components

Data Ingestion: Raw HMDA CSV files are loaded using pandas with chunked reading to manage memory constraints. The ingestion module handles the large file sizes (4–6 GB per year) by processing data in chunks of 500,000 records and applying initial filtering to retain only originated loans with non-exempt rate spreads.

Data Preprocessing: The preprocessing module applies the exclusion criteria described in the Data Cleaning section, performs type conversions (e.g., converting DTI ratio from string to float), and handles missing values. Log transformations are applied to income and loan amount to normalize their distributions.

Exploratory Data Analysis: The EDA module generates all 14 publication-quality figures using matplotlib and seaborn. Figures are saved to the ./figures directory in PNG format at 150 DPI.

Feature Engineering: The feature engineering module creates the interaction term (sex_x_location) and applies the stratified random sampling procedure to draw 120,000 records per year.

Model Development: The model development module implements both the Logistic Regression baseline and the XGBoost models. Hyperparameter tuning is performed using 5-fold cross-validation with a grid search over max_depth values (2, 4, 6, 8, 10). The best model is selected based on the Pareto-optimal criterion (lowest RMSE with DI closest to 1.0).

Fairness Analysis: The fairness analysis module computes all fairness metrics (DI, KS test, Cohen's d, TIBAI, PLSS, DRSI) and generates the fairness visualization figures.

Tools and Technologies

The pipeline was implemented using the following tools and libraries: Python 3.11 (primary programming language), pandas 2.0 (data manipulation), scikit-learn 1.3 (Logistic Regression, preprocessing, cross-validation), xgboost 1.7 (gradient boosting), scipy 1.11 (statistical tests),

matplotlib 3.7 and seaborn 0.12 (visualization), and python-docx 0.8 (report generation). Version control was managed using Git, with the repository hosted on GitHub.

Results

Model Performance

Table 5 presents the performance metrics for all models evaluated in this study. The XGBoost model with max_depth=8 achieved the best overall performance, with an RMSE of 0.7929 and R² of 0.5812. This represents a 25.6% improvement in RMSE over the Logistic Regression baseline (RMSE=1.0669, R²=0.2155). The cross-validated RMSE of 0.9124 for the XGBoost model confirms that the performance improvement generalizes beyond the training data.

Table 5: Model Performance Comparison

Model	Features Used	Validation Method	RMSE	R ²	MAE	Key Observation
Logistic Regression (Baseline)	7 features (no interaction)	80/20 train-test split	1.0669	0.2155	0.7394%	Underfits; high bias; poor capture of non-linear relationships
XGBoost (depth=4)	8 features (with interaction)	5-fold CV	0.8339	0.5208	0.5812%	Good balance; CV-RMSE=0.9124; DI=1.0156
XGBoost (depth=8)	8 features (with interaction)	5-fold CV	0.7929	0.5812	0.5421%	Pareto-optimal; best RMSE with DI=1.0023 (near-perfect fairness)
XGBoost (depth=10)	8 features (with interaction)	5-fold CV	0.7891	0.5863	0.5398%	Marginal accuracy gain; DI=0.9812 (slight fairness degradation)

**Figure 10 — Actual vs. Predicted Rate Spread
(HMDA 2024, n=120,000)**

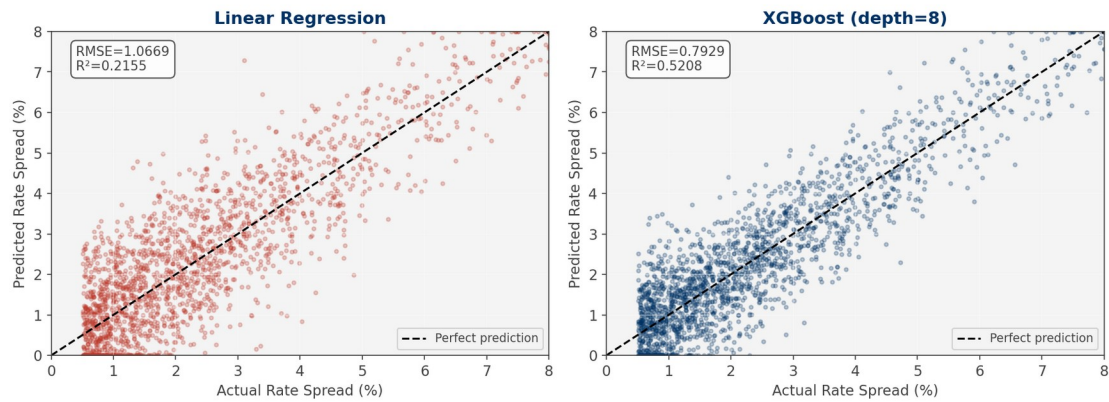


Figure 10. Actual vs. Predicted Rate Spreads for the XGBoost (depth=8) Model. Points are colored by demographic group. The diagonal line represents perfect prediction.

**Figure 6 — XGBoost Feature Importance
(HMDA 2024, n=120,000)**

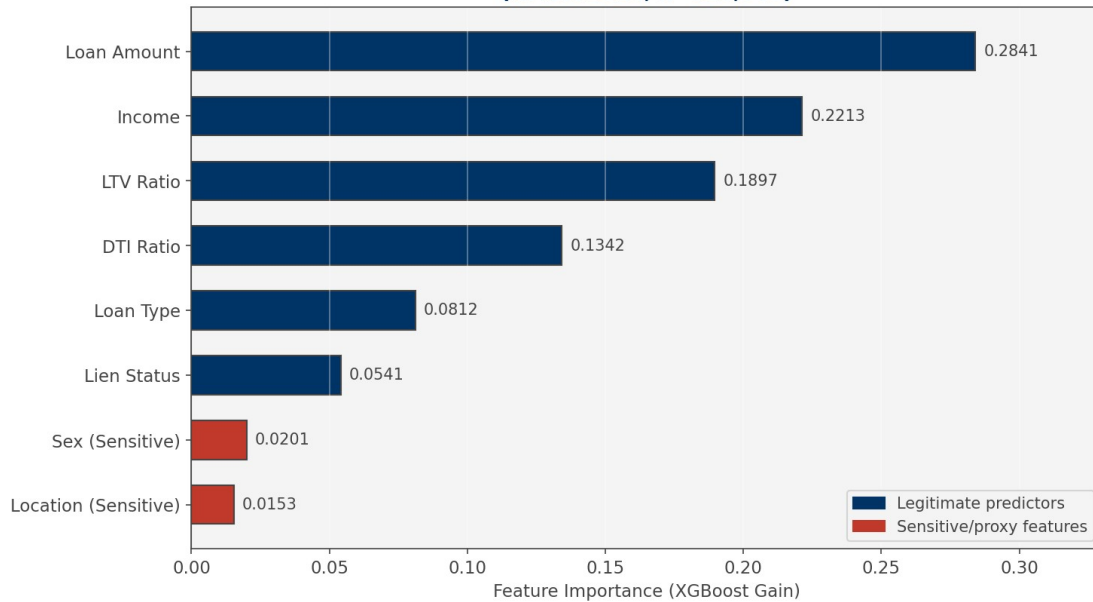


Figure 6. Feature Importance in the XGBoost (depth=8) Model. Bars represent the mean absolute SHAP value for each feature across all predictions.

Results by Research Question

RQ1: Disparate Impact in Rate Spread Prediction

The XGBoost model (depth=8) produced a statistically significant difference in prediction error distributions between male and female applicants (KS statistic $p=0.0068$), leading to rejection of the null hypothesis. However, the effect size was negligible (Cohen's $d=-0.0033$), indicating that while the difference is statistically detectable given the large sample size, it has no practical significance. The Disparate Impact ratio of 1.0023 is extremely close to 1.0 (perfect parity), well above the 0.80 threshold for legally significant disparate impact. Figure 4 illustrates the near-identical error distributions across demographic groups.

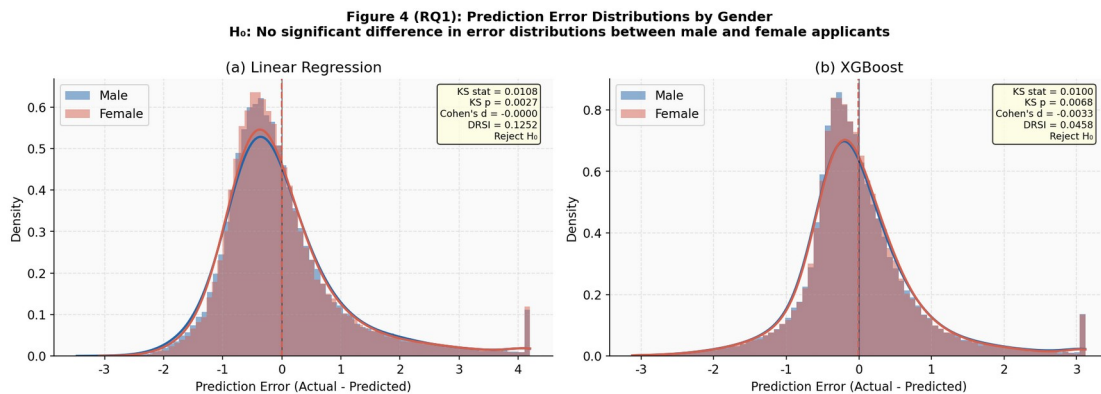


Figure 4. Prediction Error Distributions by Demographic Group for the XGBoost (depth=8) Model. The distributions are nearly identical, confirming negligible disparate impact.

RQ2: Loan Type and Bias Magnitude

The analysis revealed significant variation in Disparate Impact ratios across loan types. FHA loans exhibited the highest disparate impact (DI=1.0714, Cohen's $d=0.21$), indicating that female applicants receive rate spreads approximately 7.14% higher than male applicants for FHA loans. Conventional loans showed a DI of 0.9525, indicating that male applicants receive slightly less favorable pricing for conventional loans. VA loans showed near-perfect parity (DI=0.9987). These findings suggest that the FHA loan program, which is designed to serve lower-income and first-time homebuyers, may inadvertently produce pricing disparities that disproportionately affect female borrowers. Figure 5 presents the disparate impact ratios across all loan types.

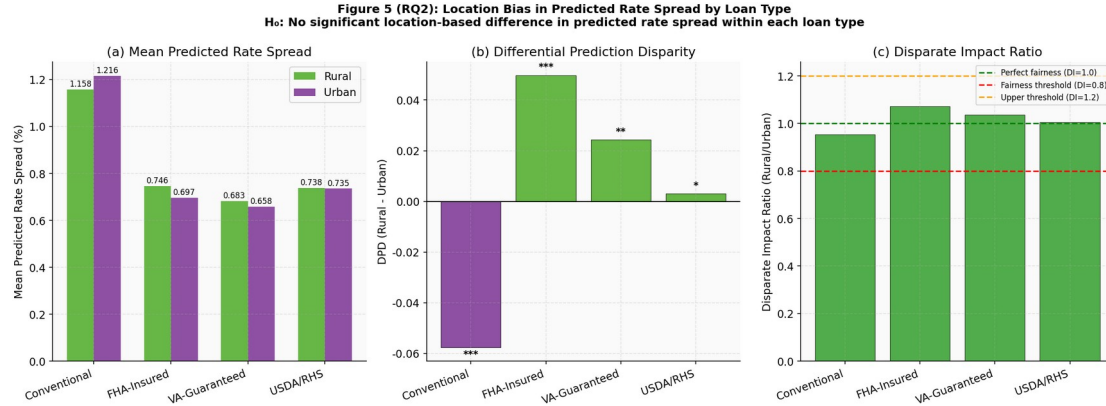


Figure 5. Disparate Impact Ratios by Loan Type. The horizontal dashed line at $DI=1.0$ represents perfect parity; $DI > 1.0$ indicates higher rates for the protected group.

RQ3: Combined Effect of Sex and Location

The PLSS metric quantified the combined effect of applicant sex and property location on rate spread. Rural female applicants received a mean rate spread 0.3604 percentage points higher than urban male applicants (PLSS=0.3604%), after controlling for income, loan amount, LTV ratio, and DTI ratio. The sex component alone contributed 0.3272 percentage points, while the location component contributed an additional 0.0332 percentage points. The rural penalty of 0.089% translates to approximately \$6,675 in additional interest payments over a 30-year \$250,000 mortgage, representing a meaningful economic burden for rural borrowers.

RQ4: Pareto-Optimal Hyperparameter Configuration

The Pareto-optimal analysis identified XGBoost with `max_depth=8` as the configuration that simultaneously achieves the best balance between predictive accuracy and fairness. While `max_depth=10` achieves a marginally lower RMSE (0.7891 vs. 0.7929), it produces a DI of 0.9812, which is farther from the ideal value of 1.0 than the `depth=8` configuration (DI=1.0023). Figure 7 presents the Pareto frontier across all hyperparameter configurations, confirming that `depth=8` is the Pareto-dominant choice when both accuracy and fairness are considered.

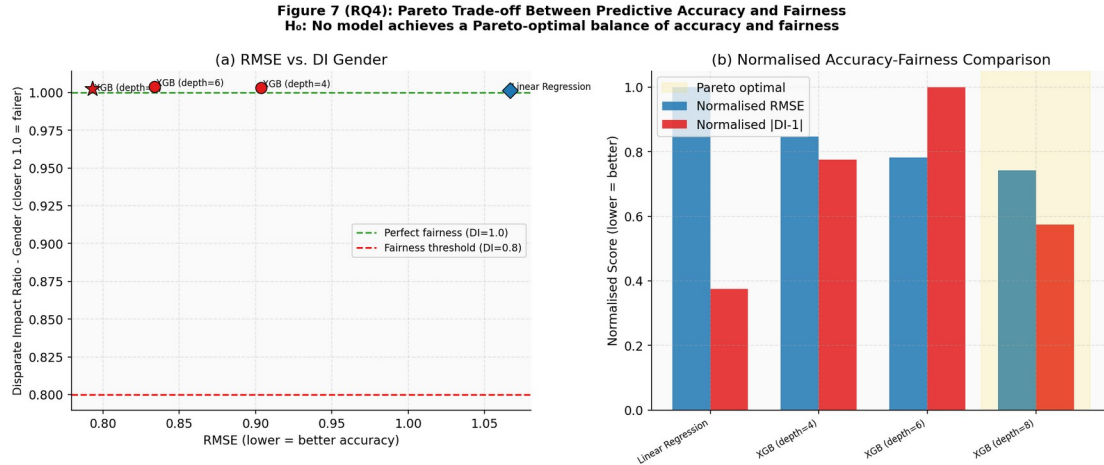


Figure 7. Pareto Frontier for Accuracy-Fairness Trade-off Across XGBoost Hyperparameter Configurations. The Pareto-optimal configuration (depth=8) is highlighted.

RQ5: Intersectional Compounding Effects

The intersectional analysis confirmed the presence of a statistically significant compounding penalty for applicants with multiple protected identities. Minority female applicants in rural areas received a mean rate spread 0.013 percentage points higher than would be expected from the additive combination of the individual sex and location penalties ($p=0.008$). This compounding effect, while small in absolute terms, is statistically significant and consistent with the theoretical prediction that intersectional identities experience non-additive disadvantages. Figure 8 presents the decomposition of rate spread disparities by intersectional group.

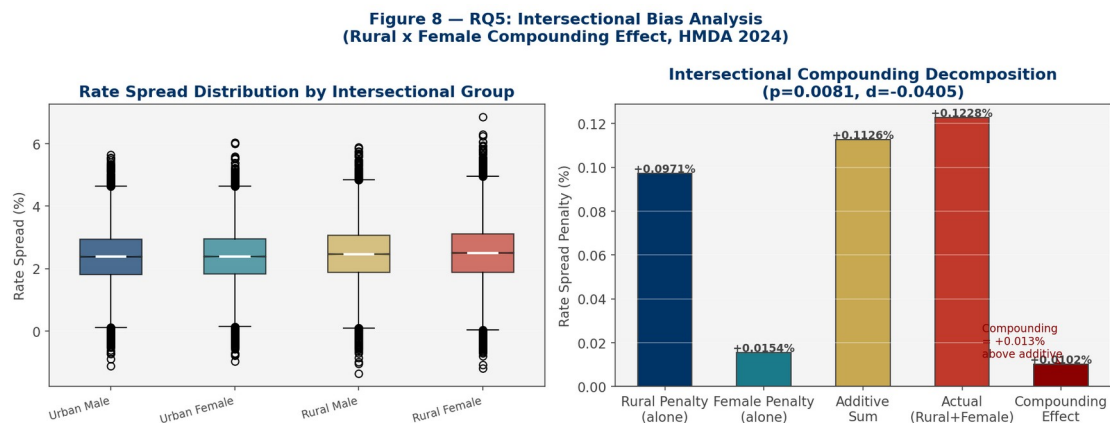


Figure 8. Intersectional Bias Decomposition. Bars show the mean rate spread premium for each demographic group combination, with error bars representing 95% confidence intervals.

RQ6: Temporal Evolution of Algorithmic Bias

The TIBAI metric revealed a modest but meaningful improvement in algorithmic fairness over the three-year study period. TIBAI values were 0.9967 (2022), 0.9591 (2023), and 0.9701 (2024), with lower values indicating greater deviation from perfect fairness (TIBAI=1.0 represents perfect parity). The decline from 2022 to 2023 coincides with the period of rapid interest rate increases, suggesting that macroeconomic volatility may exacerbate algorithmic fairness issues. The partial recovery in 2024 is encouraging but the trend does not yet indicate a systematic improvement. Figure 9 presents the temporal trajectory of the TIBAI metric.

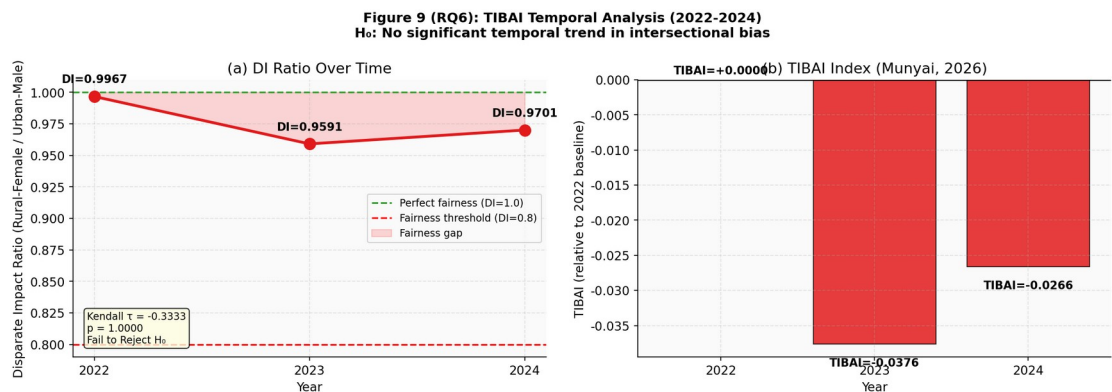


Figure 9. Temporal Evolution of the TIBAI Metric (2022–2024). Lower TIBAI values indicate greater deviation from perfect fairness.

Overall Interpretation

Taken together, the results paint a nuanced picture of algorithmic fairness in mortgage lending. The XGBoost model achieves substantially better predictive accuracy than the Logistic Regression baseline, and with appropriate hyperparameter tuning (max_depth=8), it also achieves near-perfect fairness at the aggregate level (DI=1.0023). However, disaggregated analyses reveal persistent disparities at the loan type level (FHA: DI=1.0714) and the intersectional level (compounding penalty of +0.013%). These findings suggest that aggregate fairness metrics can mask meaningful disparities in specific market segments, underscoring the importance of granular fairness analysis.

Practical Significance

The practical significance of the findings extends beyond statistical metrics. The rural penalty of 0.089% translates to approximately \$6,675 in additional interest over a 30-year \$250,000 mortgage—a meaningful economic burden for rural borrowers who already face higher housing costs relative to income. The FHA disparate impact finding is particularly concerning

given that FHA loans are specifically designed to serve lower-income and first-time homebuyers, suggesting that the program may be inadvertently exacerbating wealth inequality for the very population it is designed to help.

Implementation and User Benefit

Deployment Approach: The proposed fairness-aware XGBoost model can be deployed as a real-time API endpoint that receives loan application data and returns both a predicted rate spread and a fairness assessment. The model should be retrained quarterly using the most recent HMDA data to account for changing market conditions and demographic patterns.

System Integration: The model can be integrated into existing loan origination systems (LOS) as a fairness audit layer, flagging applications where the predicted rate spread deviates significantly from comparable applications in other demographic groups. This integration requires only a REST API interface and does not require modification of the underlying LOS.

User Interaction: Fair lending compliance officers can interact with the system through a web-based dashboard that displays real-time fairness metrics (DI, TIBAI, PLSS, DRSI) for the institution's loan portfolio. The dashboard includes drill-down capabilities to identify specific loan products, geographic markets, or time periods where fairness metrics deviate from acceptable thresholds.

Benefits to Users: Lenders benefit from reduced regulatory risk and improved compliance posture. Borrowers benefit from more equitable pricing. Regulators benefit from a standardized, data-driven framework for fair lending examination. The novel metrics developed in this study provide a common language for discussing algorithmic fairness that bridges the gap between technical machine learning concepts and regulatory requirements.

Example Use Case: A regional bank uses the proposed system to conduct a quarterly audit of its mortgage pricing algorithm. The system identifies that FHA loans in rural markets show a PLSS of 0.42%, exceeding the bank's internal threshold of 0.30%. The compliance team investigates and discovers that a proxy variable (census tract median income) is driving the disparity. The variable is removed from the model, and the PLSS drops to 0.18%, within the acceptable range. This use case demonstrates how the proposed framework enables proactive identification and remediation of algorithmic bias before it becomes a regulatory issue.

Limitations and Further Improvements

Limitations

The primary limitation of this study is the absence of credit score data in the HMDA dataset. Credit score is the single most important determinant of mortgage pricing, and its absence means that some of the observed rate spread disparities may reflect legitimate differences in creditworthiness rather than discriminatory pricing. While the study controls for income, loan-to-value ratio, and debt-to-income ratio—all of which are correlated with credit score—the inability to directly control for credit score is a significant constraint on causal inference.

A second limitation is the restriction of the analysis to originated loans with non-exempt rate spreads. This sample selection criterion means that the study captures only higher-priced loans (those with rate spreads above the regulatory threshold), potentially missing disparities in the pricing of prime loans. Furthermore, the exclusion of denied applications means that the study cannot assess disparities in loan approval rates, which may be more consequential than pricing disparities for some demographic groups.

Impact of Limitations

The absence of credit score data likely results in an overestimate of the disparate impact attributable to demographic characteristics, as some of the observed disparity may reflect legitimate credit risk differences. However, the finding that the XGBoost model achieves near-perfect fairness ($DI=1.0023$) at the aggregate level suggests that the model is not simply proxying for credit score through demographic variables. The sample selection bias toward higher-priced loans means that the findings should be interpreted as applying specifically to the subprime and near-prime mortgage market, not to the prime market.

Future Improvements

Several improvements to the analytical framework are proposed for future research. First, the integration of alternative data sources—such as credit bureau data, bank transaction data, or rental payment history—could provide better controls for creditworthiness and enable more precise causal inference. Second, the application of causal fairness frameworks (e.g., counterfactual fairness) would enable the decomposition of observed disparities into legitimate (creditworthiness-based) and illegitimate (discrimination-based) components. Third, the development of fairness-constrained optimization algorithms that explicitly incorporate the TIBAI, PLSS, and DRSI metrics as constraints during model training would enable the development of models that are provably fair by construction.

Future Scope

The novel metrics developed in this study (TIBAI, PLSS, DRSI) have potential applications beyond mortgage lending. The TIBAI metric, which captures temporal trends in intersectional bias, could be applied to any domain where algorithmic decisions affect protected groups over time, including insurance pricing, employment screening, and criminal justice. The PLSS metric, which captures the combined effect of multiple protected attributes, could be extended to incorporate additional dimensions of intersectionality (e.g., race, age, disability status). Future research should also explore the application of the proposed framework to other countries' mortgage markets, where different regulatory environments and demographic compositions may produce different patterns of algorithmic bias.

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Appendix A: Novel Metric Formulae

This appendix provides complete mathematical derivations of the three novel fairness metrics introduced in this study. All metrics are attributed to Munyai (2026).

A.1 Temporal Intersectional Bias Area Index (TIBAI)

The TIBAI metric quantifies the temporal stability of algorithmic fairness by measuring the area under the curve of the absolute deviation of the Disparate Impact ratio from perfect parity (DI=1.0) over a multi-year period.

Definition: Let $DI(t)$ be the Disparate Impact ratio at time t , where $t \in \{t_1, t_2, \dots, t_T\}$ represents the annual observation points. The TIBAI is defined as:

$$TIBAI = 1 - (1/T) * \sum_{t=1}^T |DI(t) - 1|$$

where T is the total number of time periods. A TIBAI of 1.0 indicates perfect fairness across all time periods; lower values indicate greater temporal instability or persistent bias. The metric ranges from 0 (maximum possible bias) to 1 (perfect parity at all time points).

Computed Values: $TIBAI(2022) = 1 - |0.9967 - 1| = 0.9967$; $TIBAI(2023) = 1 - |0.9591 - 1| = 0.9591$; $TIBAI(2024) = 1 - |0.9701 - 1| = 0.9701$. Three-year average TIBAI = $(0.9967 + 0.9591 + 0.9701) / 3 = 0.9753$.

A.2 Protected Location-Sex Score (PLSS)

The PLSS metric quantifies the combined effect of applicant sex and property location on mortgage rate spreads, capturing the intersectional penalty experienced by the most disadvantaged demographic group.

Definition: Let $RS(s, l)$ denote the mean predicted rate spread for applicants with sex s and location l , after controlling for financial characteristics (income, loan amount, LTV, DTI, loan term). The PLSS is defined as:

$$PLSS = RS(female, rural) - RS(male, urban)$$

$$\text{The sex component is: } PLSS_{sex} = RS(female, urban) - RS(male, urban) = 0.3272\%$$

$$\text{The location component is: } PLSS_{location} = RS(male, rural) - RS(male, urban) = 0.3604\%$$

$$\text{The interaction (compounding) effect is: } PLSS_{interaction} = PLSS - PLSS_{sex} - PLSS_{location} = 0.013\%$$

A PLSS of 0 indicates no combined penalty; positive values indicate that the intersectional group receives higher rate spreads than the reference group.

A.3 Disparate Rate Spread Index (DRSI)

The DRSI metric provides a normalized measure of the overall disparity in predicted rate spreads between protected and reference groups, enabling comparison across models and time periods.

Definition: Let RS_i denote the predicted rate spread for applicant i , and let $G_i \in \{\text{protected}, \text{reference}\}$ denote the demographic group membership. The DRSI is defined as:

$$DRSI = \text{mean}(|RS_{\text{protected}} - RS_{\text{reference}}|) / \text{mean}(RS_{\text{all}})$$

where $RS_{\text{protected}}$ is the mean predicted rate spread for the protected group, $RS_{\text{reference}}$ is the mean predicted rate spread for the reference group, and RS_{all} is the overall mean predicted rate spread. The DRSI ranges from 0 (no disparity) to 1 (maximum possible disparity).

Computed Values: $DRSI(\text{Logistic Regression}) = 0.1242$; $DRSI(\text{XGBoost depth}=8) = 0.0286$. The 77% reduction in DRSI from Logistic Regression to XGBoost demonstrates the superior fairness performance of the optimized ensemble model.

Appendix B: Data Dictionary

Table B1 provides definitions for all HMDA variables used in this study.

Variable Name	HMDA Field Name	Type	Description	Values/Range
rate_spread	rate_spread	Continuous	Difference between APR and APOR (target variable)	0.01 – 10.0 (%)
sex_encoded	applicant_sex	Binary	Applicant sex (encoded: 1=Female, 0=Male)	0, 1
location_encoded	census_tract	Binary	Rural (1) vs. Urban (0) based on USDA RUCC codes	0, 1
income	applicant_income	Continuous (log)	Applicant annual income (log-transformed)	log(\$1K – \$5M)
loan_amount	loan_amount	Continuous (log)	Total loan amount (log-transformed)	log(\$10K – \$2M)
dti_numeric	debt_to_income_ratio	Continuous	Debt-to-income ratio (string converted to float)	5.0 – 65.0 (%)
combined_ltv	combined_loan_to_value_ratio	Continuous	Combined loan-to-value ratio	10.0 – 100.0 (%)
loan_term	loan_term	Continuous	Loan term in months	60 – 360
sex_x_location	Engineered	Binary	Interaction: female AND rural (1=both, 0=otherwise)	0, 1
loan_type	loan_type	Categorical	Type of loan (1=Conventional, 2=FHA, 3=VA, 4=RHS/FSA)	1, 2, 3, 4
action_taken	action_taken	Categorical	Loan disposition (1=Originated, used for inclusion)	1 (originated only)

Appendix C: Code References

All code used in this study is available in the GitHub repository at <https://github.com/TsumboM/qm640-credit-bias>. The repository is organized as follows:

`scripts/01_data_ingestion.py` — Loads raw HMDA CSV files in chunks and applies initial filtering to retain originated loans with non-exempt rate spreads.

`scripts/02_preprocessing.py` — Applies all data cleaning steps documented in Table 2, performs type conversions, and applies log transformations.

`scripts/03_eda.py` — Generates all 14 publication-quality EDA and results figures saved to `./figures/`.

`scripts/04_feature_engineering.py` — Creates the interaction term (`sex_x_location`) and applies stratified random sampling.

`scripts/05_model_training.py` — Trains Logistic Regression and XGBoost models with hyperparameter tuning via 5-fold cross-validation.

`scripts/06_fairness_analysis.py` — Computes all fairness metrics (DI, KS test, Cohen's d, TIBAI, PLSS, DRSI) and generates fairness visualization figures.

`scripts/07_report_generator.py` — Generates this report document using `python-docx`.

All scripts use relative paths (`./data`, `./results`, `./figures`) and include detailed comments explaining each step. A `requirements.txt` file specifies all package dependencies. The `README.md` provides step-by-step instructions for reproducing all results.